



Fig. 6: Connections between ESC, flight controller, and remote receiver



Fig. 7: Control buttons on the flight controller

receiver and hence the drone. Binding steps and configurations are normally explained in the user manual.

Construction and assembly

First screw in the four arms of quadcopter to the central frame as shown in Fig. 3. Solder the ESC to the frame power distribution circuit board. Mount the LiPo battery on the middle of drone frame. You may use insulation tape for this.

Now mount the KK 2.2 flight controller in the middle of the frame and fix it with adhesive as shown in Fig. 4. Next fix the motors on the arms with screws and attach all four ESCs to the arms. Attach the correct propellers to the motors.

ESC calibration

We need to calibrate all the ESCs so that all the motors start at the same time and get the top speed on full throttle. Most flight controllers have settings of

their own for ESC calibration. Manual calibration for each ESC can be done using radio receiver and transmitter.

For calibration, connect a motor to ESC and then to channel 3 (throttle of RC receiver). Keep the throttle knob at maximum position on radio transmitter and turn on the transmitter. After that connect the LiPo battery and power on the ESC. Wait for two beeps and then move the throttle stick/knob to minimum position and wait for a few seconds till you hear three beeps. Then unplug and plugin the LiPo battery. Slowly move the throttle stick and the motors will start to rotate. With this, one of your ESCs is calibrated.

Repeat the same steps with all other ESCs one by one to calibrate them. The assembled quadcopter drone with ESC module mounted on each arm is shown in Fig. 5.

One end of each ESC has three wires for motor connections. The other end of ESC has five cables: two thick cables for power supply (black pin for GND and red pin for positive) and three thin cables for +5V supply, signal, and ground pins, respectively. Yellow cable is the signal pin for radio receiver's channel 3.

On transmitter side, keep the throttle stick to low position and turn it on by connecting to the LiPo battery. (These steps require removal of all the propellers attached to the motors.) Then push the throttle stick slowly until you see the motor moving clockwise or anticlockwise. If it is clockwise, the motor should be at front left. Then swipe/interchange the two wires of ESC and check the reverse direction of motor. Repeat the same process with all the motors and make sure that the rotations of the motors are as per the directions shown in Fig. 2.

Now you may attach the propellers to the motors.

Connect yellow wire of each ESC to signal pin of flight controller. The connections between the ESC, flight controller, and remote receiver are shown in Fig. 6. Connect ESC1 through ESC4 to first row through fourth row pins of flight controller, respectively. Again, connect first row through fourth row pins of flight controller to first column through fourth column pins of remote receiver, respectively.

Flight controller settings

Next, we need to do some settings in the flight controller. First put the throttle stick down to minimum



Fig. 8: Accelerometer calibration